Week-14 Assessment: Configuring VNet-to-VNet VPN in Azure

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1. Assessment Objective

The objective of this lab is to connect two isolated Azure Virtual Networks (**VNet-1** and **VNet-2**) using a Site-to-Site style VPN (**VNet-to-VNet**). This creates a secure, encrypted IPSec tunnel over the Microsoft backbone, allowing resources in different networks to communicate using private IP addresses.

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2. Core Concepts

2.1 The GatewaySubnet

The `GatewaySubnet` is a dedicated subnet reserved exclusively for Azure VPN Gateway instances.

- **Purpose:** Azure deploys managed gateway VMs into this subnet to handle encryption and routing.
- **Constraint:** It must be named exactly **"GatewaySubnet"**.
- **Constraint:** You cannot deploy your own Virtual Machines (VMs) in this subnet.

2.2 VPN Gateway & Public IP

The VPN Gateway acts as the "doorway" between networks. Although the traffic between VNets remains within the Azure infrastructure, the gateways use Public IPs to establish the encrypted IPSec/IKE tunnel.

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3. Architecture Overview

- **VNet-1 (10.0.0.0/16):** Contains `Subnet1` (for VM1) and `GatewaySubnet` (for VPN Gateway 1).
- **VNet-2 (10.1.0.0/16):** Contains `Subnet1` (for VM2) and `GatewaySubnet` (for VPN Gateway 2).
- **The Tunnel:** A secure VNet-to-VNet connection linking Gateway 1 and Gateway 2.

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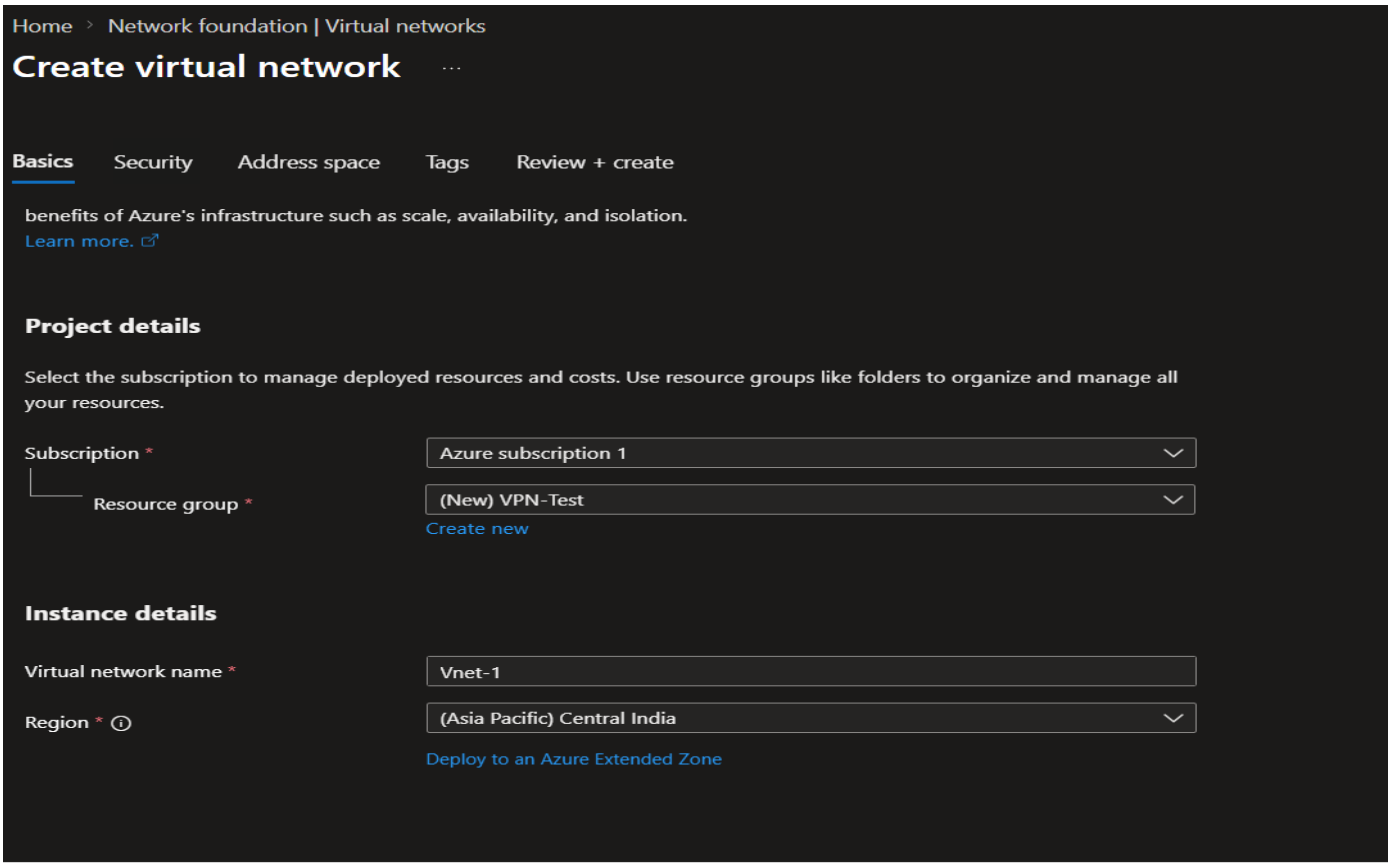
4. Step-by-Step Implementation

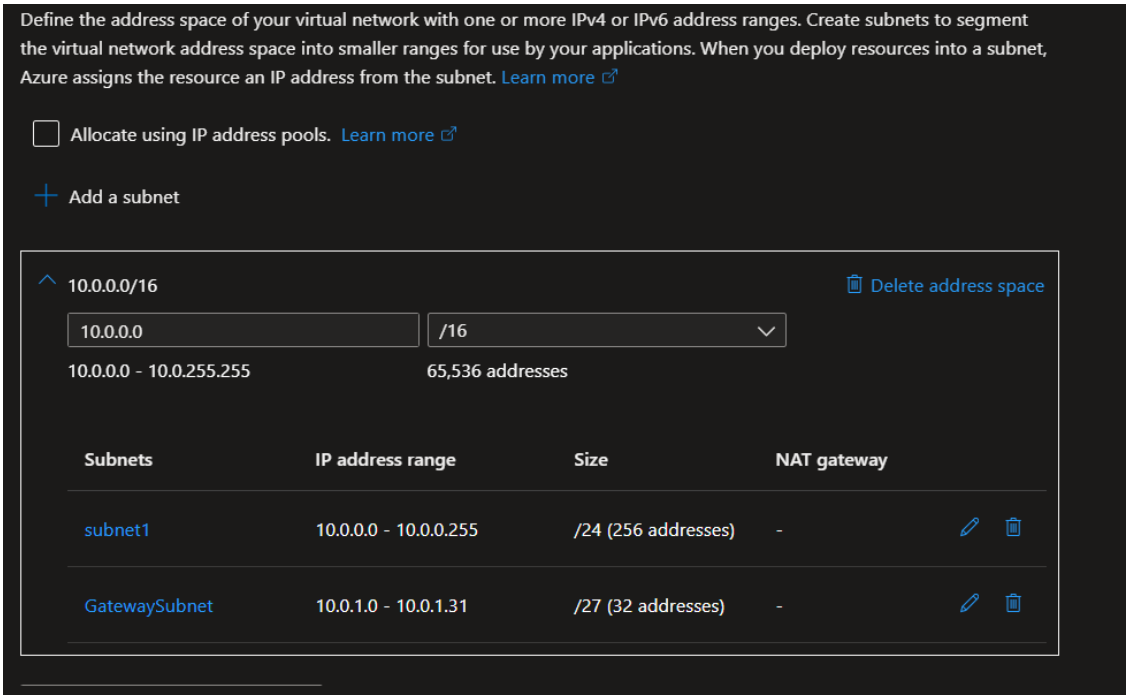
Step 1: Resource Group Creation

- **Action:** Created a resource group named `rg-vnet-vpn-lab`.
- **Region:** Central India (or your preferred region).

Step 2: Virtual Network Configuration

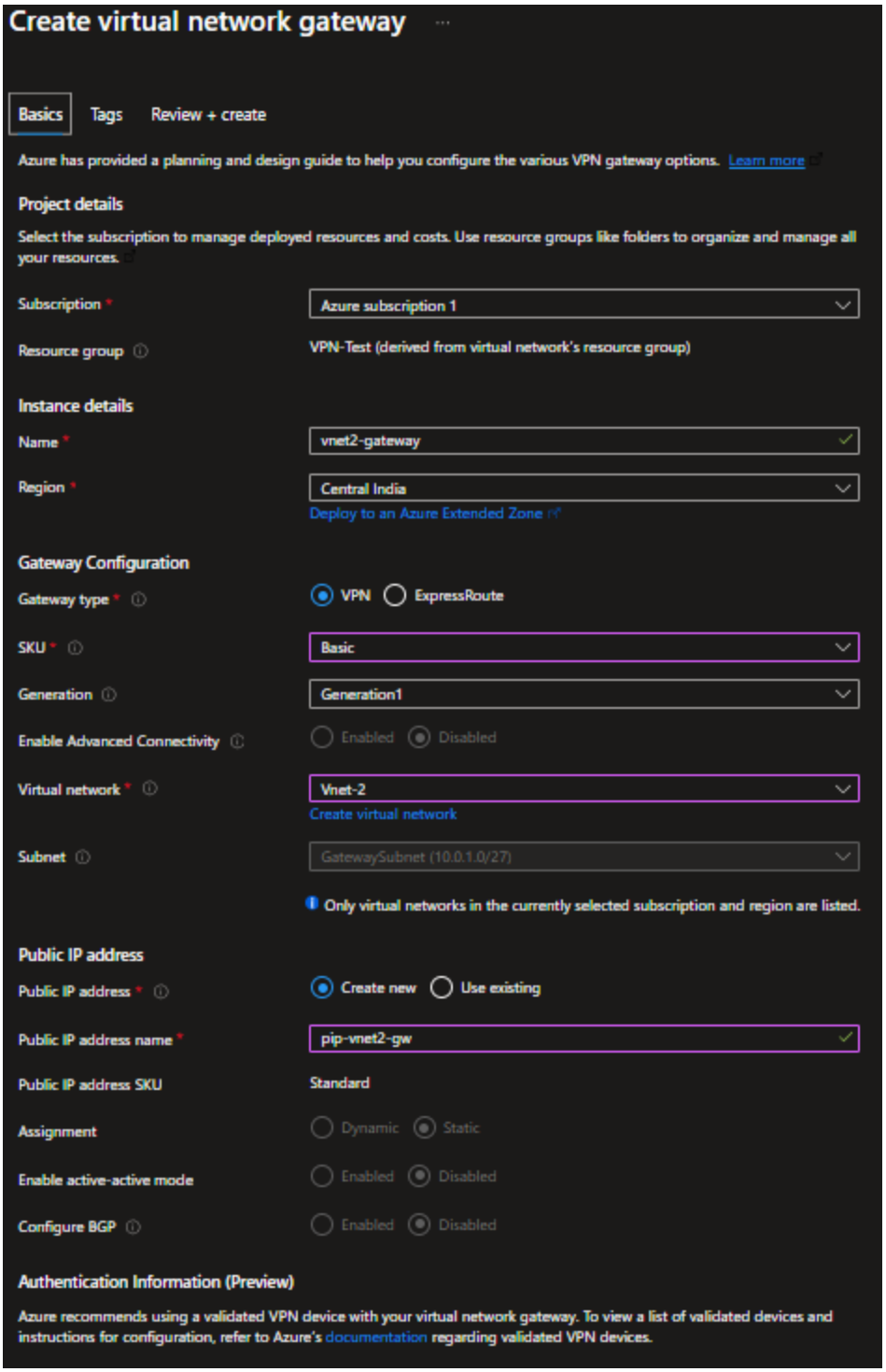
- **VNet-1 Setup:**
  - **Address Space:** `10.0.0.0/16`
  - **Workload Subnet:** `subnet1` (`10.0.1.0/24`)
  - **Gateway Subnet:** `GatewaySubnet` (`10.0.255.0/27`)
- **VNet-2 Setup:**
  - **Address Space:** `10.1.0.0/16`
  - **Workload Subnet:** `subnet1` (`10.1.1.0/24`)
  - **Gateway Subnet:** `GatewaySubnet` (`10.1.255.0/27`)





Step 3: Deploying Virtual Network Gateways

- **Gateway 1 (vnet1-gateway):**
  - **Type:** VPN | **VPN Type:** Route-based | **SKU:** VpnGw1.
  - **Public IP:** Created `pip-vnet1-gw`.
- **Gateway 2 (vnet2-gateway):**
  - **Type:** VPN | **VPN Type:** Route-based | **SKU:** VpnGw1.
  - **Public IP:** Created `pip-vnet2-gw`.
- **Note:** Deployment took approximately 25 minutes per gateway.



Step 4: Establishing Bi-Directional Connections

To enable communication, two connection objects were created using a **Shared Key (PSK)**: `Azure123`.

1. **Connection 1:** `vnet1-to-vnet2` (Source: `vnet1-gateway` -> Destination: `vnet2-gateway`).
2. **Connection 2:** `vnet2-to-vnet1` (Source: `vnet2-gateway` -> Destination: `vnet1-gateway`).

Step 5: Deployment of Test Workloads

- **VM1:** Deployed in `vnet1/subnet1`.
- **VM2:** Deployed in `vnet2/subnet1`.

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5. Verification and Connectivity Test

Test Results:

- **Source:** VM1 (Private IP: `10.0.1.x`)
- **Target:** VM2 (Private IP: `10.1.1.x`)
- **Command:** `ping 10.1.1.x`
- **Outcome:** The ping was successful, confirming that the VPN tunnel is encrypting and routing traffic between the two isolated VNets.

```
azureuser@VM1:~$ ping 10.1.0.4
PING 10.1.0.4 (10.1.0.4) 56(84) bytes of data.
64 bytes from 10.1.0.4: icmp_seq=1 ttl=64 time=9.25 ms
64 bytes from 10.1.0.4: icmp_seq=2 ttl=64 time=9.22 ms
64 bytes from 10.1.0.4: icmp_seq=3 ttl=64 time=9.55 ms
64 bytes from 10.1.0.4: icmp_seq=4 ttl=64 time=61.4 ms
^C
--- 10.1.0.4 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 9.222/22.355/61.396/22.540 ms
azureuser@VM1:~$ nano vmkey.pem
azureuser@VM1:~$ chmod 400 vmkey.pem
azureuser@VM1:~$ ssh azureuser@10.1.0.4 -i vmkey.pem
```

```
azureuser@VM2:~$ ping 10.0.0.4
PING 10.0.0.4 (10.0.0.4) 56(84) bytes of data.
64 bytes from 10.0.0.4: icmp_seq=1 ttl=64 time=6.05 ms
64 bytes from 10.0.0.4: icmp_seq=2 ttl=64 time=4.19 ms
64 bytes from 10.0.0.4: icmp_seq=3 ttl=64 time=3.53 ms
64 bytes from 10.0.0.4: icmp_seq=4 ttl=64 time=6.05 ms
64 bytes from 10.0.0.4: icmp_seq=5 ttl=64 time=4.09 ms
64 bytes from 10.0.0.4: icmp_seq=6 ttl=64 time=3.40 ms
64 bytes from 10.0.0.4: icmp_seq=7 ttl=64 time=60.5 ms
64 bytes from 10.0.0.4: icmp_seq=8 ttl=64 time=3.55 ms
64 bytes from 10.0.0.4: icmp_seq=9 ttl=64 time=4.25 ms
^C
--- 10.0.0.4 ping statistics ---
9 packets transmitted, 9 received, 0% packet loss, time 8011ms
rtt min/avg/max/mdev = 3.398/10.624/60.525/17.667 ms
```

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6. Conclusion

The lab successfully demonstrated the deployment of a VNet-to-VNet VPN. By utilizing the `GatewaySubnet`, we provided the necessary environment for Azure's managed gateway VMs to perform secure encryption (IPSec) and cross-network routing. This architecture ensures that sensitive data remains private while moving between different cloud environments.